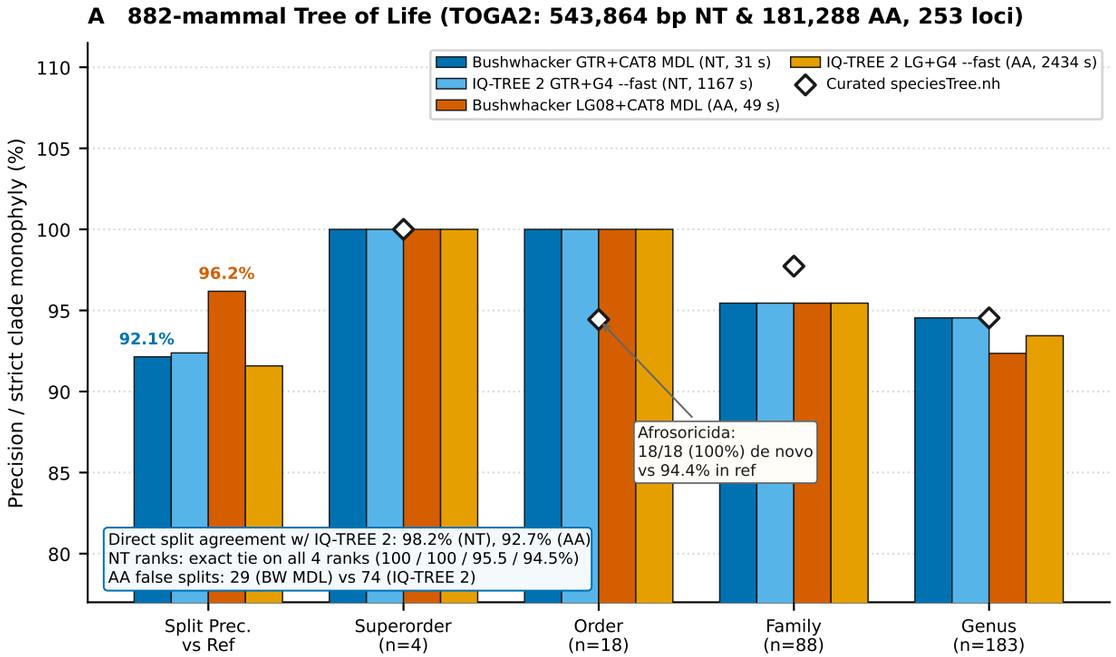
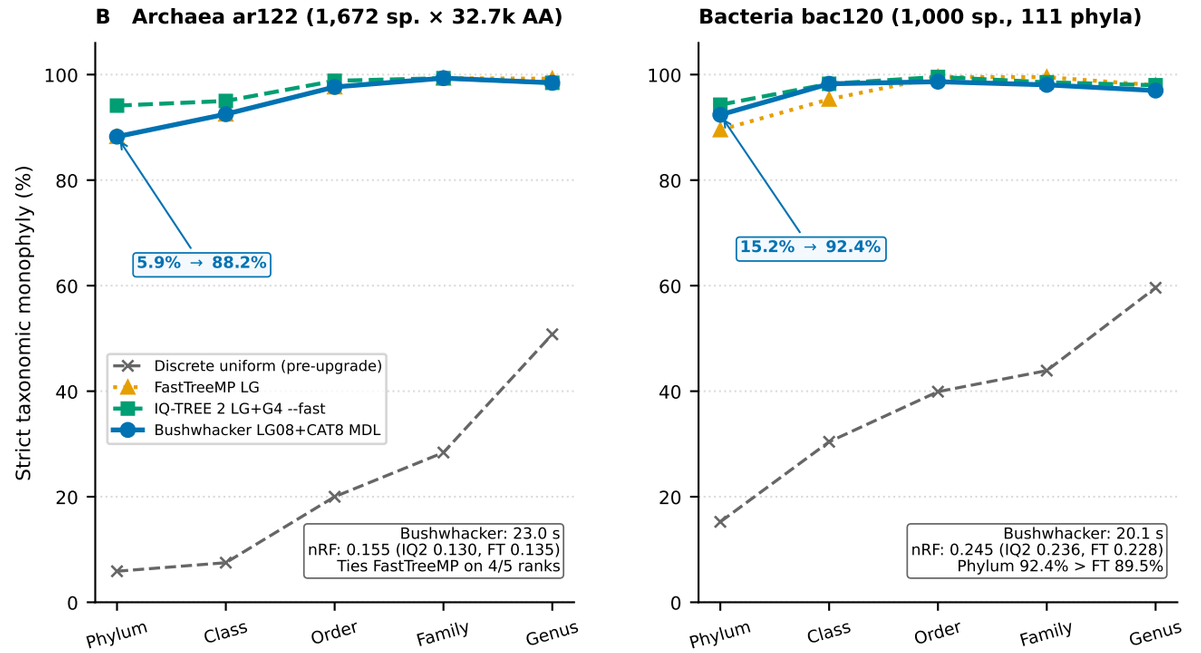
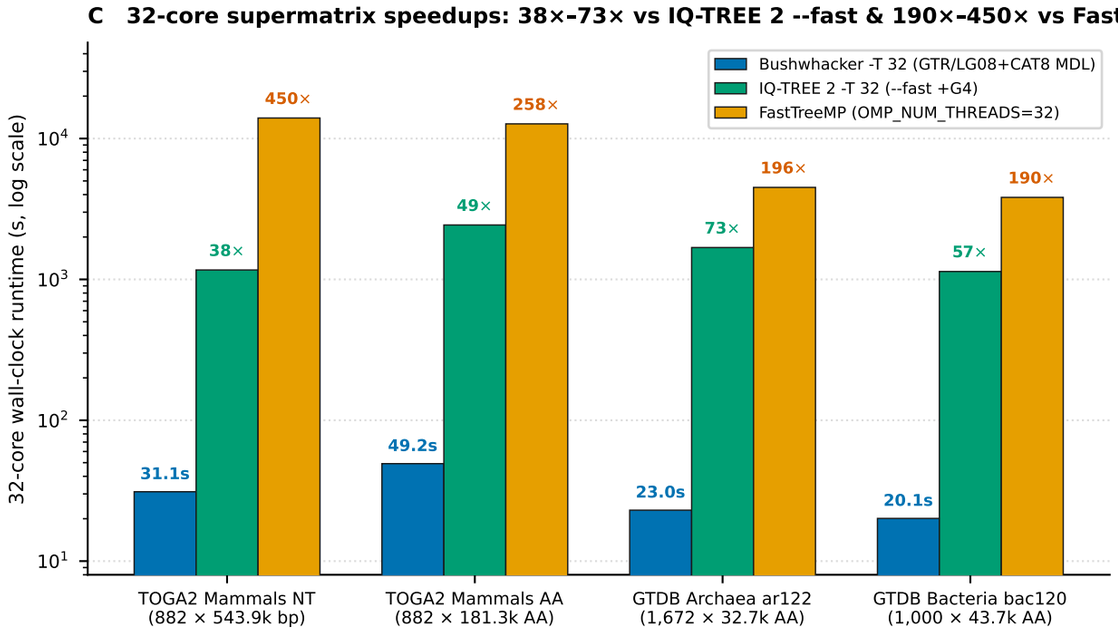
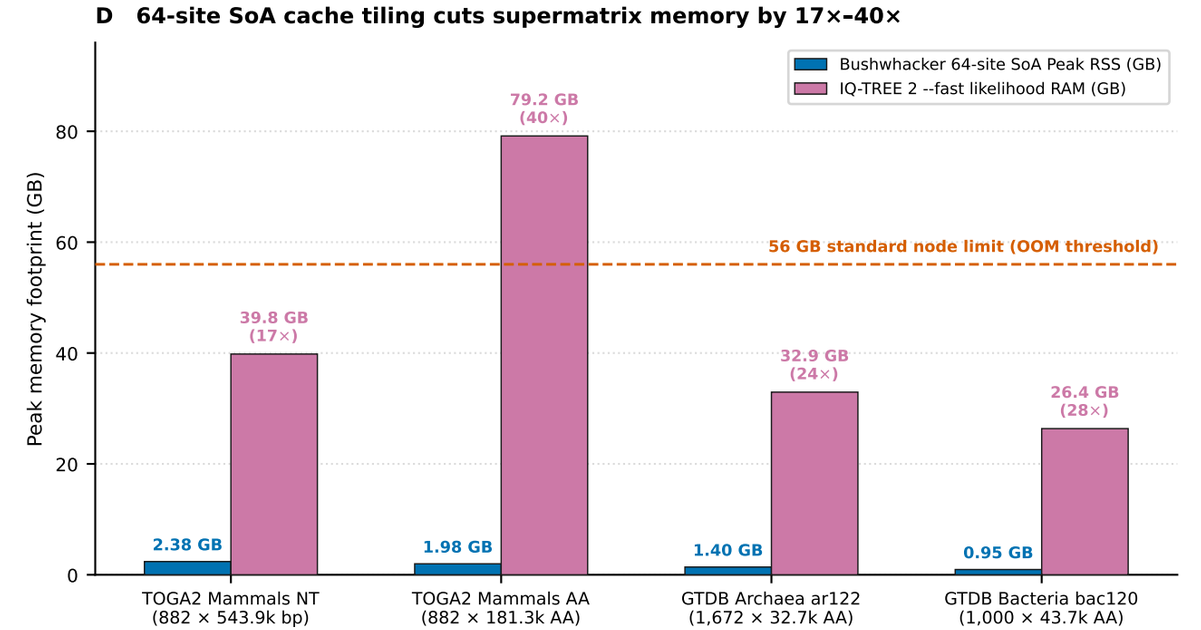

A 882-mammal Tree of Life (TOGA2: 543,864 bp NT & 181,288 AA, 253 loci)
Bushwhacker GTR+CAT8 MDL (NT, 31 s)
IQ-TREE 2 GTR+G4 --fast (NT, 1167 s)
Bushwhacker LG08+CAT8 MDL (AA, 49 s)
IQ-TREE 2 LG+G4 --fast (AA, 2434 s)
Curated speciesTree.nh
Precision / strict clade monophyly (%)
92.1%
96.2%
Afrosoricida:
18/18 (100%) de novo
vs 94.4% in ref
Direct split agreement w/ IQ-TREE 2: 98.2% (NT), 92.7% (AA)
NT ranks: exact tie on all 4 ranks (100 / 100 / 95.5 / 94.5%)
AA false splits: 29 (BW MDL) vs 74 (IQ-TREE 2)
Split Prec. vs Ref
Superorder (n=4)
Order (n=18)
Family (n=88)
Genus (n=183)


B Archaea ar122 (1,672 sp. × 32.7k AA)
Bacteria bac120 (1,000 sp., 111 phyla)
Strict taxonomic monophyly (%)
5.9% → 88.2%
15.2% → 92.4%
Discrete uniform (pre-upgrade)
FastTreeMP LG
IQ-TREE 2 LG+G4 --fast
Bushwhacker LG08+CAT8 MDL
Bushwhacker: 23.0 s
nRF: 0.155 (IQ2 0.130, FT 0.135)
Ties FastTreeMP on 4/5 ranks
Bushwhacker: 20.1 s
nRF: 0.245 (IQ2 0.236, FT 0.228)
Phylum 92.4% > FT 89.5%
Phylum
Class
Order
Family
Genus


C 32-core supermatrix speedups: 38×–73× vs IQ-TREE 2 --fast & 190×–450× vs FastTreeMP
Bushwhacker -T 32 (GTR/LG08+CAT8 MDL)
IQ-TREE 2 -T 32 (--fast +G4)
FastTreeMP (OMP_NUM_THREADS=32)
32-core wall-clock runtime (s, log scale)
31.1s
38×
450×
49.2s
49×
258×
23.0s
73×
196×
20.1s
57×
190×
TOGA2 Mammals NT
(882 × 543.9k bp)
TOGA2 Mammals AA
(882 × 181.3k AA)
GTDB Archaea ar122
(1,672 × 32.7k AA)
GTDB Bacteria bac120
(1,000 × 43.7k AA)


D 64-site SoA cache tiling cuts supermatrix memory by 17×–40×
Bushwhacker 64-site SoA Peak RSS (GB)
IQ-TREE 2 --fast likelihood RAM (GB)
Peak memory footprint (GB)
56 GB standard node limit (OOM threshold)
2.38 GB
39.8 GB
(17×)
1.98 GB
79.2 GB
(40×)
1.40 GB
32.9 GB
(24×)
0.95 GB
26.4 GB
(28×)
TOGA2 Mammals NT
(882 × 543.9k bp)
TOGA2 Mammals AA
(882 × 181.3k AA)
GTDB Archaea ar122
(1,672 × 32.7k AA)
GTDB Bacteria bac120
(1,000 × 43.7k AA)